Supporting Information

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SI Methods

Coculture System. *R. pomeroyi* DSS-3 was streaked to isolation on 1/2 YTSS agar and grown at 30°C in the dark for 2 d, after which isolated colonies were inoculated into liquid 1/2 YTSS at 30°C and grown overnight in the dark with shaking. Cells were harvested in exponential phase (~0.5 optical density, OD) by centrifugation for 5 min at 4,500 × *g* and washed twice in f/2 medium (+Si, $-B_{12}$) (1). Cells were resuspended in f/2 (+Si, $-B_{12}$) to ~1×10^8 cells mL^{-1} and a dilution of 1:100 was used for inoculation. *T. pseudonana* CCMP1335 was cultured in f/2 medium (+Si) in a diurnal incubator at 18°C on a 16:8 h light:dark cycle at ~160 μmol photons m^{-2} s^{-1}. Axenic status was verified regularly using 4,6-diamidino-2-phenylindole (DAPI) staining and epifluorescence microscopy along with plating on 1/2 YTSS agar. *T. pseudonana* was limited for vitamin B_{12} by sequential transfer in f/2 without B_{12} (+Si, $-B_{12}$) until cell growth rate decreased significantly (~50%) compared with the vitamin. B_{12}-limited *T. pseudonana* was diluted to ~10^4 cells mL^{-1} in fresh f/2 (+Si, $-B_{12}$).

Four treatments were established ($n = 2$ or 4): (*i*) *T. pseudonana* + B_{12} (369 pM cyanocobalamin), (*ii*) *T. pseudonana* − B_{12}, (*iii*) *T. pseudonana* + *R. pomeroyi*, and (*iv*) *R. pomeroyi* alone. Growth rates of *T. pseudonana* in the presence of *R. pomeroyi* DSS-3 cells compared with exogenous B_{12} (0.78 d^{-1} and 0.75 d^{-1}, respectively) were not statistically different from one another (ANCOVA, $F = 0.74$, ref. 2). Bacterial abundance was tracked with a Beckman Coulter CyAn flow cytometer using 0.5 mL volumes that were fixed with glutaraldehyde (2% vol/vol final concentration) and stored at −80°C until analysis. The thawed samples were stained with SYBR-Green II (Molecular Probes/Invitrogen) for 10 min on ice in the dark, and cells were enumerated using a combination of side-scatter and green fluorescence detection by comparison with fluorescent beads (Flouresbrite TG Carboxylate 1.75 μm Microspheres; Polysciences) added in known quantities. Diatom abundance was tracked by relative fluorescent units (RFU) on a BioTek Synergy Mx microplate reader and/or enumerated via flow cytometry as described above for bacterial cells except using a combination of side-scatter and red fluorescence detection. For RT-qPCR studies of expression of DHPS pathway gene *hpsN*, two additional treatments were established in which *R. pomeroyi* DSS-3 was grown in f/2 (+Si, $-B_{12}$) with 5 mM acetate or 3.33 mM DHPS as the sole carbon source (both at 10 mM carbon). DHPS was kindly provided by K. Denger and A. Cook (Konstanz, Germany).

Transcriptome Analysis. Cells were collected for RNA at the midpoint of the light cycle by filtration onto 47 mm diameter 0.2-μm pore-size Supor or polycarbonate filters after 8 h (550 mL, bacterial transcriptomics) or during exponential phase (350 mL, diatom transcriptomics; 250 mL, bacterial RT-qPCR). Filters were flash frozen in liquid N_2 and subsequently stored at −80 °C. RNA was prepared for sequencing as previously described for environmental metatranscriptomic samples (3, 4) with some modifications. In short, filters were broken into small pieces and transferred to 50 mL conical tubes containing 4 mL of Ambion lysis buffer (AM8540G) plus 0.5 mL of Sigma glass beads (G8772). Tubes were vortexed for 10 min, and RNA was purified using an RNeasy Kit (Qiagen). Total RNA was treated for DNA contamination using two successive treatments with Turbo-DNase (Ambion; Life Technologies) and cleaned/concentrated with an RNeasy MinElute Cleanup kit (Qiagen). For the bacterial RNA samples, ribosomal RNAs (rRNAs) were selectively removed using a subtractive hybridization protocol (5) with biotinylated rRNA probes specific to the organism(s) in each sample (e.g., 16S and 23S for *R. pomeroyi* DSS-3, 18S and 28S for *T. pseudonana*) and polyA-tailed RNAs were removed using a biotinylated oligo dT probe. Probe-bound RNAs were removed with strepdavidin-coated magnetic beads (New England Biolabs). Depleted samples were linearly amplified using the MessageAmp II-Bacteria kit (Ambion; Life Technologies). In the case of the diatom RNA samples, polyA-tailed RNAs were isolated from total RNA using the Poly(A)Purist Kit (Ambion; Life Technologies), and mRNAs were linearly amplified using the MessageAmp II aRNA kit (Ambion; Life Technologies).

Amplified mRNAs were converted into cDNAs for either Illumina or SOLiD sequencing. cDNAs for Illumina sequencing were synthesized using the SuperScript III First-Strand Synthesis System (Invitrogen; Life Technologies) followed by the NEBNext mRNA Second Strand Synthesis Module (New England Biolabs). cDNAs were then purified using the QIAquick PCR purification kit (Qiagen) followed by EtOH precipitation. SOLiD cDNAs were prepared with the SOLiD Whole Transcriptome Analysis kit (Applied Biosystems).

Next-generation Sequencing. For Illumina sequencing, cDNA was sheared ultrasonically to ~250-bp fragments and libraries were constructed for single-end (1 × 50) sequencing using the Illumina MiSeq sequencing platform (Illumina).

For SOLiD sequencing, emulsion PCR was performed on cDNA samples using the SOLiD EZ Bead System (Life Technologies), and sequencing was performed with a Life Technologies SOLiD sequencer version 4.

Differential Transcriptome Analysis. Sequenced reads were mapped to the genomes of *R. pomeroyi* DSS-3 (6) and *T. pseudonana* (Thaps3 FilteredModels2) (7) using BWA (v.0.5.9) (8). The number of sequence reads that aligned to gene models were calculated from the resulting SAM alignment files, and only sequences aligning to gene models were used for subsequent analyses.

Mapped read counts were analyzed for significant differential expression (FDR < 0.05) using the baySeq package in R (9). For the diatom transcriptomes, a two-way comparison was performed on axenic ($n = 2$) versus bacterized ($n = 2$) replicates; for the bacterial transcriptomes, a two-way comparison was performed on replicates with ($n = 4$) versus without ($n = 2$) diatoms. A gene was designated as significantly differentially expressed if it met the FDR cutoff in at least two of the three independent analyses, to control for random effects of subsampling.

RT-qPCR. Primers were designed to amplify a 148-bp region of the *hpsN* gene (SPO0594) using Primer3 (10). All RT-qPCRs were run in triplicate in 25-μL volumes on DNased total RNA with the iScript One-Step RT-PCR Kit with SYBR Green (Bio-Rad) with specific priming cDNA reverse transcription. Each reaction contained 300 nM final primer concentration and between 1.0 μL and 3.0 μL of RNA template. All reactions were run on an iCycler iQ (Bio-Rad) with the following cycling conditions: initial 10 min 50 °C reverse transcription step, denaturation at 95 °C for 5 min, 40 cycles of 95 °C for 30 s followed by annealing at 51 °C for 30 s, and then a final denaturation and annealing for 1 min followed by a melt curve. Copies per reaction were normalized to total ng of input RNA in each reaction, quantified via the Quant-iT RiboGreen RNA Reagent and Kit (Invitrogen; Life Technologies).

A standard for *hpsN* was constructed with PCR gene products amplified from *R. pomeroyi* DSS-3 genomic DNA. The PCR product was cleaned (QIAquick PCR purification kit; Qiagen) and checked on an agarose gel to verify it was a single amplicon of the correct size. The amplicon was inserted into a PCR-4 vector and cloned into *E. coli* using a TA-TOPO kit (Life Technologies). The plasmid was extracted (QIAquick Spin Miniprep kit; Qiagen) and the cloned region sequenced to verify the correct insert. The plasmid was linearized with NotI, and after cleanup (QIAquick PCR purification kit; Qiagen), quantified via the Quant-iT PicoGreen dsDNA Reagent and Kit (Invitrogen; Life Technologies). A 10-fold serially diluted standard was run on each plate.

Identification of DHPS Genes in Reference Bacterial Genomes. The full-length protein sequence for HpsN in *R. pomeroyi* DSS-3, other roseobacters, and other bacterial taxa for which function has been experimentally verified (11) were used as queries to identify orthologs in bacterial genomes available in IMG-ER and Roseobase (www.roseobase.org) using BLASTp. Full-length reference sequences and potential ortholog sequences were aligned using MAFFT (12) with the L-ISN-i algorithm and the Blosum62 matrix. A high-quality alignment region consisting of 400 amino acids was selected using the Gblocks program (13). The alignment file was converted to Phylip format using Readseq (iubio.bio.indiana.edu/cgi-bin/readseq.cgi), and the best-fit amino acid substitution matrix (LG) and among-site rate heterogeneity model (gamma distribution) were determined using the ProtTest software (14). Outgroups included a variety of closely related histidinol dehydrogenase sequences.

Because the substantial variation of genomic composition (e.g., G+C content) across taxa included in the tree might affect HpsN composition, model adequacy with respect to amino acid composition was assessed using the χ^2 test on posterior distributed samples, generated using the posterior predictive simulation implemented in the P4 Bayesian phylogenetic software package (15). The model comparison was performed using the Bayes factor (16), and marginal likelihoods were estimated using equation 16 in Newton and Raftery (17) based on the harmonic mean of the likelihoods from posterior distributions. The χ^2 test showed that a Bayesian composition-homogeneous model was adequate despite substantial variation in genomic composition among the taxa, and a tree was constructed in P4 software with a composition-homogeneous model. A tree was also constructed using the node-discrete composition heterogeneity (NDCH) model in P4, to allow for different composition on different branches. For this, Bayesian Metropolis-coupled Markov Chain Monte Carlo (MCMC) analyses using different configurations of the NDCH model were performed. The amino acid sequences were analyzed using the LG matrix plus Gamma distribution model and by using the polytomy prior (18). Five independent replicate runs were performed for each configuration of the NDCH model, with one cold and three heated MCMC chains run for a total of 2,500,000 generations, with trees sampled every 1,000 generations and the first 750,000 generations discarded as "burn-in." An average SD of split support of <0.1 suggested that convergence was reached, and a majority-rule consensus tree was constructed from the postburn-in trees. A tree was also constructed that accounted for the possibility that different domains of the protein were under distinct selective pressure, using the empirical mixture model C60 implemented in the PhyloBayes software package (19) under the C60-LG model with a Gamma distribution of rates among sites. Four independent MCMC runs were performed, each with >60,000 cycles, and the first 20% of all runs were discarded as "burn-in." Convergence was reached with the maxdiff statistic of 0.05.

Targeted Diatom Metabolite Analysis. Cells were collected from 100 mL of culture during the midpoint of the light cycle by centrifugation at 2,880 × *g* for 5 min. Pellets were flash frozen in liquid N_2 and stored at −80 °C. Frozen cells were suspended in 1.3 mL HPLC grade methanol at 4 °C and extraction was allowed to proceed at −20 °C for 20 min. Samples were centrifuged for 5 min (16.1 rcf) at 4 °C. The supernatant was transferred to new vials and two additional pellet extractions were carried out by resuspending in 50 μL of 80% methanol in water at 4°C followed by extraction at −20 °C for 20 min. Vials containing the combined supernatants were dried under N_2 until the solvent evaporated. Solid residue was resuspended in 300 μL of sterile water and kept at 4°C.

A 10-μL aliquot was injected through a Synergi 2.5 μm Hydro-RP 100, 100 × 2.00 mm LC column (Phenomenex) kept at 25 °C. The ultra-performance mass spectrometer was run in full scan mode with negative ionization using a method adapted from Rabinowitz (20). The eluent was introduced into the MS via an electrospray ionization source conjoined to a Thermo Scientific Exactive Plus orbitrap mass spectrometer through a 0.1 mm internal diameter fused silica capillary tube. The samples were run with a spray voltage of 3 kV. The nitrogen sheath gas was set to a flow rate of 10 psi with a capillary temperature of 320 °C and an Automatic Gain Control target set to 3×10^6. The samples were analyzed with a resolution of 140,000 and a scan window of 85–800 *m/z* from 0 to 9 min and 110–1000 *m/z* from 9 to 25 min. The chromatography gradient was: initial hold of 100% solvent A (97:3 water:methanol, 10 mM tributylamine, and 15 mM acetic acid 0.1% formic acid in water):0% B (methanol) for 5 min, ramp to 20% B from 5 to 13 min, ramp to 55% B from 13 to 15.5 min, ramp to 85% B from 15.5 to 19 min, and ramp to 0% B from 19 to 25 min, with a flow rate of 200 μL min^{-1}.

Raw data files generated by Xcalibur (Thermo Scientific) were converted to the open source mzML format (21) via the ProteoWizard package (22). Maven (23, 24) was used to automatically correct the total ion chromatograms based on the retention times for each sample. Known metabolites were manually selected by mass (±5 ppm) and retention time.

Targeted Community Metabolite Analysis. Plankton (>1.6 μm, 5 m depth) were collected from four stations during the GeoMICS project Line P transect (Stations P1, P4, P6, and P8) in May 2012. Plankton cells were caught on a 1.6 μm pore-size filter (Whatman GF/A). Filters were frozen at −80 °C until processing. Immediately before extraction, plankton filters were removed from the freezer and sectioned, and subsamples were placed in preweighed combusted glass amber vial. Metabolite extraction was adapted from Rabinowitz and Kimball (25). Filters were sonicated in 500 μL of cold 40:40:20 acetonitrile:methanol:0.1% formic acid in water for 10 min to lyse the cells, and extracts transferred to Eppendorf centrifuge tubes and centrifuged at 17,000 × *g* for 5 min. The supernatant was transferred to a fresh amber vial, neutralized with 6 M ammonium hydroxide, dried using vacuum centrifugation, and redissolved in 500 μL of 95:5 water:acetonitrile. Five μL of 5 μg mL^{-1} biotin-(*ring*-6,6-d_2) was added to each sample as an injection standard. Samples were kept at 4 °C until analysis on the mass spectrometer, within 24 h of extraction. Extraction blanks were conducted with Milli-Q water filtered through 1.6-μm filters.

Dissolved metabolites (collected concurrently with the filter fractions) were extracted from seawater using PPL resin, after acidification of the 0.2-μm filtrate with concentrated HCl (to pH 3), according to previously published methods (26). Samples were eluted with methanol. A fraction of each DOM extract (3 mL) was dried using nitrogen gas and reconstituted in 500 of μL 95:5 water:acetonitrile with the biotin injection standard.

Targeted metabolite concentrations were determined with a Synergi 4u Fusion – RP 80A 150 × 2.00 mm column coupled

to a Thermo Scientific TSQ Vantage Triple Stage Quadrupole Mass Spectrometer. The chromatography gradient was: initial hold of 95% A (0.1% formic acid in water):5% B (0.1% formic acid in acetonitrile) for 2 min, ramp to 65% B from 2 to 20 min, ramp to 100% B from 20 to 25 min, and hold until 32.5 min, with a 7-min reequilibration with solvent A between samples. Metabolites were quantified using multiple reaction monitoring (MRM) mode with optimal parameters determined from infusion of authentic standards. Seven-point external calibration curves (1, 5, 10, 50, 100, 500, and 1,000 ng mL^{-1}) were generated for each compound by plotting peak area against concentration. A total of 30 metabolites were quantified, and abundance of DHPS and DMSP was calculated as a percent of these measured metabolites.

Metatranscriptome Analysis. Metatranscriptomic data from plankton at GeoMICS project Line P stations P1, P4, P6, and P8 were obtained as above except that cells were collected by sequential filtration through a 2.0-μm pore-size filter (142 mm diameter polycarbonate) and a 0.2-μm pore-size filter (142 mm diameter Supor) following prefiltration through a 53-μm pore-size filter. Cells were lysed in 10 mL of Ambion lysis buffer (AM8540G) + 0.5 mL each of 0.5- and 0.1-mm zirconia beads (BioSpec Products), and an internal standard 1,006 nt in length (27) was added to the lysis buffer in known copy numbers. rRNAs were selectively removed using a subtractive hybridization method (5) with biotinylated rRNA probes specific to the environmental samples (i.e., bacterial and archaeal 16S and 23S and eukaryotic 18S and 28S). cDNAs were sheared ultrasonically to ∼250-bp fragments, and HiSeq libraries (Illumina) were constructed for paired-end (2 × 150) sequencing using the Illumina HiSEq 2500 sequencing platform (Illumina). Reads were joined using PANDAsEq (28), and paired reads were trimmed using FASTX-Toolkit (hannonlab.cshl.edu/fastx_toolkit/). rRNA, tRNA, and internal standard sequences were removed using a BLASTn search against a database containing representative rRNA and tRNA sequences along with the internal standard sequence (bit score cutoff ≥ 50). Reads representing Roseobacter-clade *hpsN* transcripts were identified using a BLASTx search against reference sequences from Roseobacter genomes. Paralogs having sequences most similar to *hpsN* were also identified. A bit score cutoff of ≥40 was used to identify Roseobacter-like *hpsN* reads in the GeoMICS libraries, and reanalysis of the positive reads against the RefSeq protein database was used to identify functional and taxonomic annotations. Transcript concentration in seawater was calculated based on recovery of the internal standard reads.

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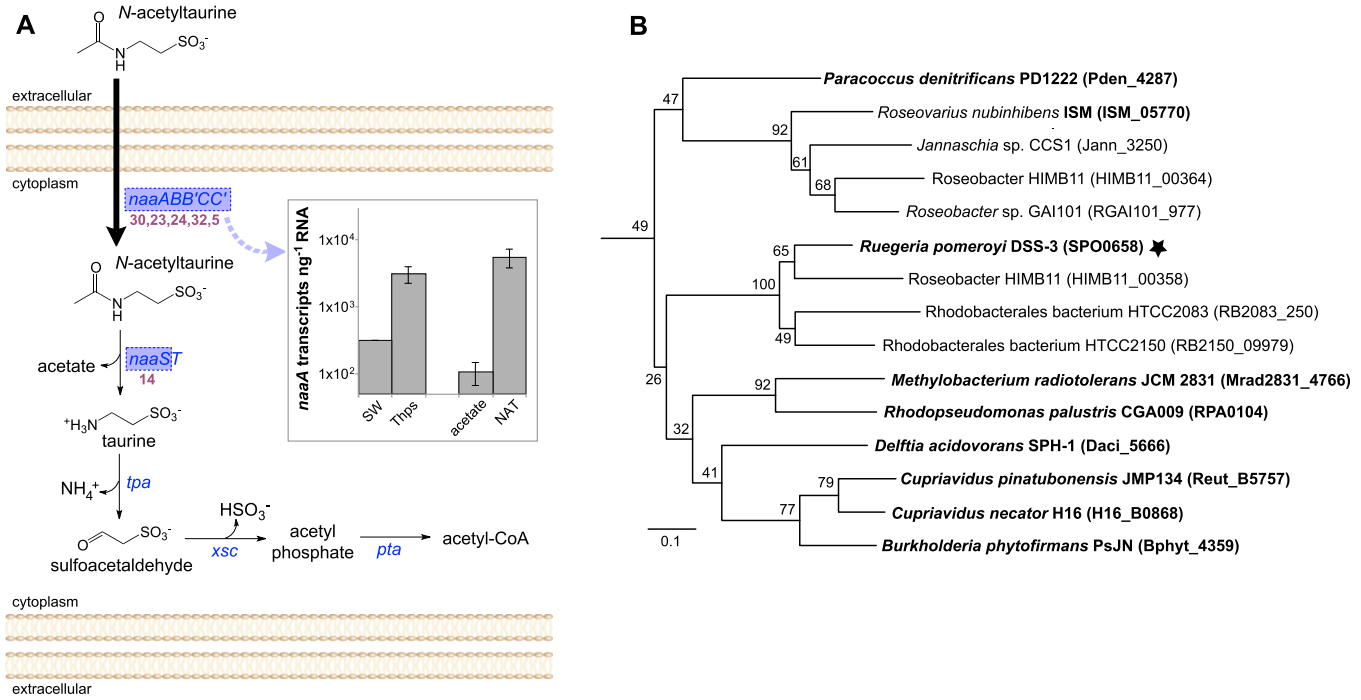


Fig. S1. *N*-acetyltaurine cycling between marine bacteria and diatoms. (*A*) *N*-acetyltaurine transport and degradation genes in the marine roseobacter *Ruegeria pomeroyi* DSS-3 (blue font) up-regulated during cogrowth with *Thalassiosira pseudonana* (purple boxes; fold-change up-regulation indicated underneath). Bold arrows indicate transporters. (*Inset*) RT-qPCR quantification of *naaA* transcripts in *R. pomeroyi* DSS-3 in coculture with *T. pseudonana* (Thps; see Fig. 1, filled circles) and in sterile f/2 seawater medium (SW) after 8 h, and during exponential growth with acetate or *N*-acetyltaurine (NAT) as the sole carbon source. Error bars represent SD of three technical replicates from duplicate cultures. (*B*) A maximum-likelihood tree derived from reference NaaS sequences, including experimentally verified sequences (bold; ref. 1) and orthologs found in marine genomes available through the Integrated Microbial Genomes database (img.jgi.doe.gov), using the RAxML method (2). Outgroups included a variety of amidohydrolase sequences. Values at the nodes show the number of times the clade appeared in 100 bootstrapped data sets, and the scale bar indicates number of substitutions per site. BLASTP analysis suggests that *N*-acetyltaurine degradation in marine bacteria may be limited to the Roseobacter clade with once exception, *Oceanibaculum indicum* P24, which belongs to the sister group Rhodospirillales. The *R. pomeroyi* DSS-3 NaaS sequence is marked by a star. All analyses were conducted the same as for DHPS and *hpsN* except that RT-qPCR primers targeted *naaA* and RAxML was used to build the NaaS tree.

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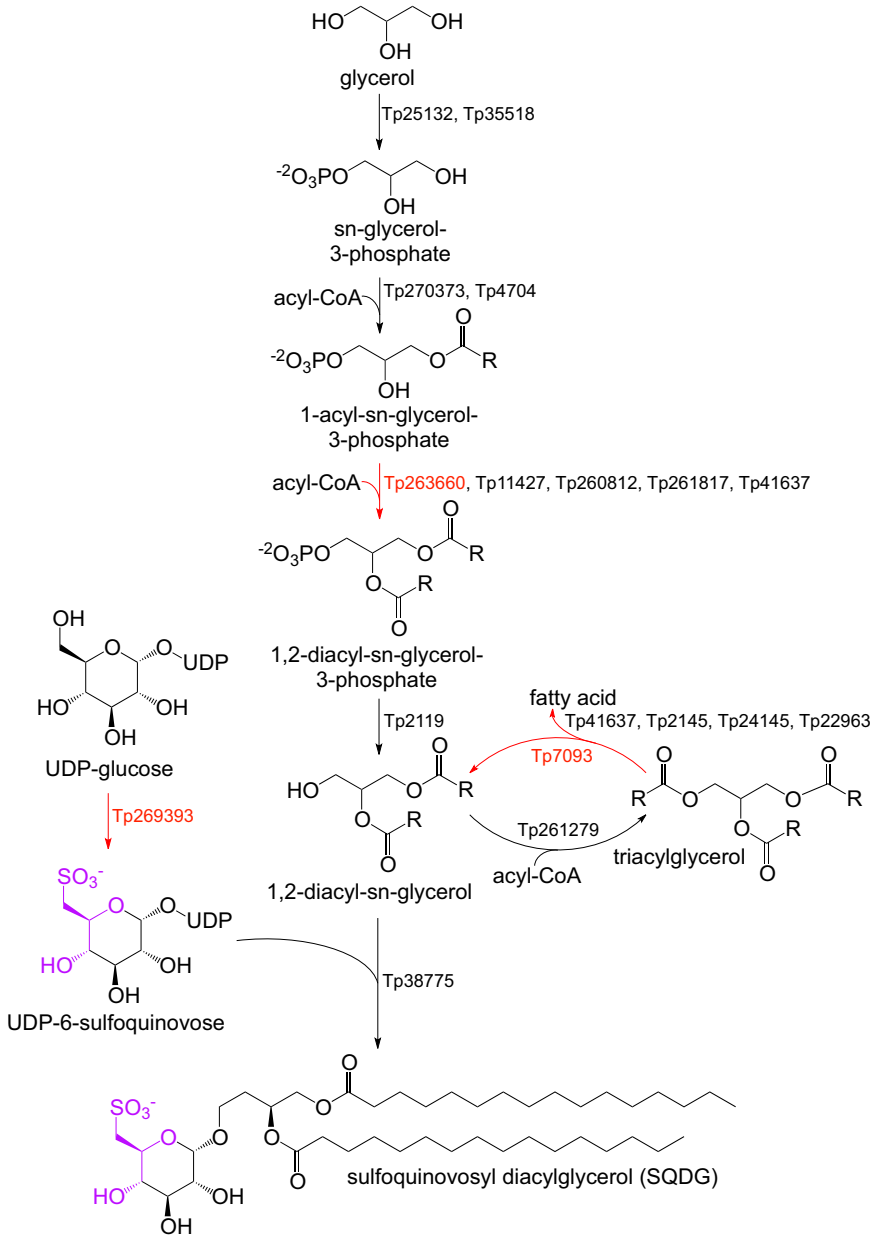


Fig. S2. Transcriptional changes in glycerolipid metabolism in *T. pseudonana*. Red arrows indicate genes down-regulated in *T. pseudonana* when cocultured with *R. pomeroyi* DSS-3 compared with axenic controls. Locus tags for unregulated homologs in the *T. pseudonana* genome are also listed. DHPS (purple) can be produced by bacteria that degrade the sulfoquinovose headgroup of SQDG (1).

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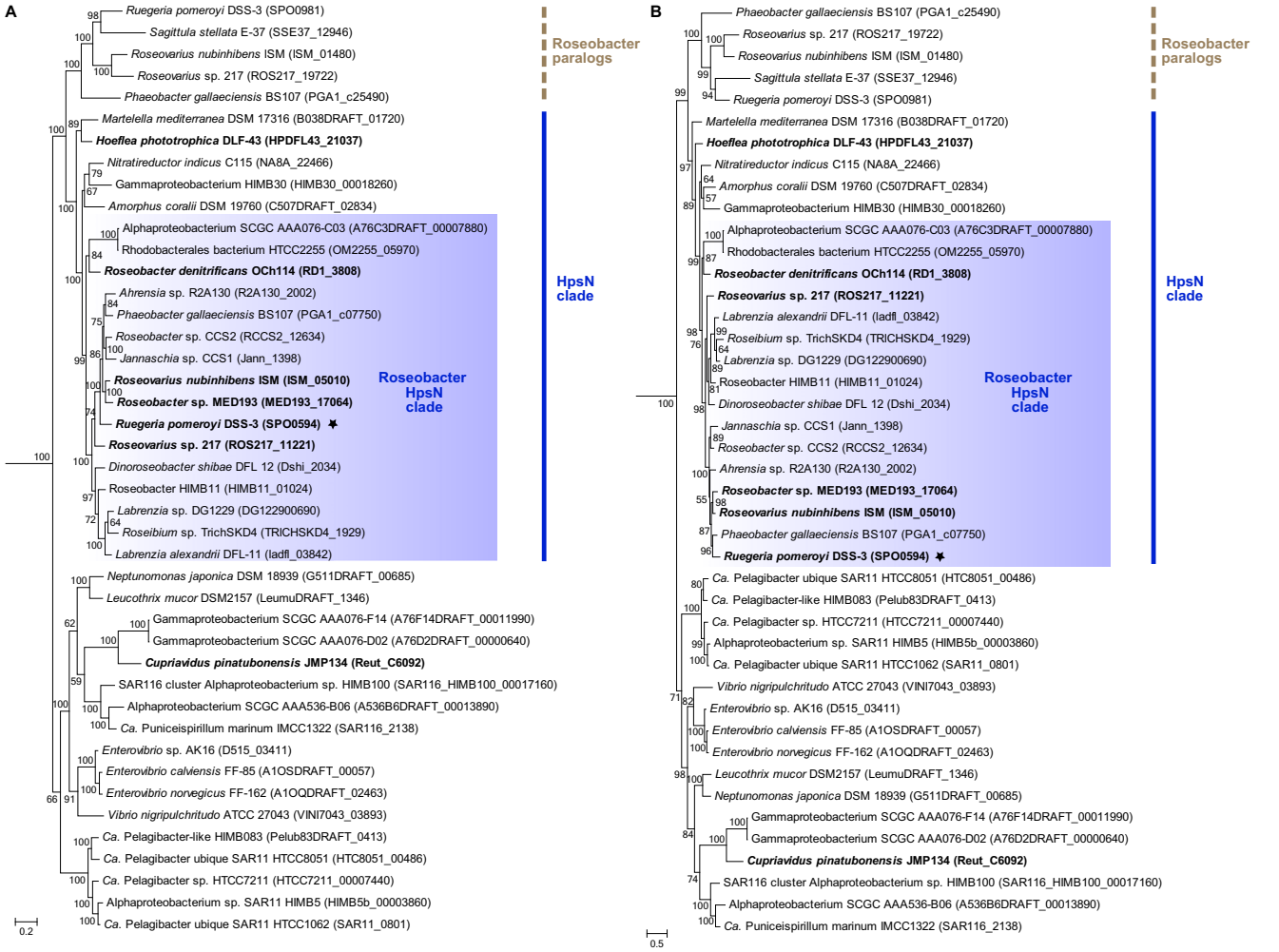


Fig. S3. Bayesian phylogenetic trees of HpsN homologs using the P4 software with a 50% majority-rule consensus tree of 1,500 trees sampled from the posterior distribution of a Metropolis-coupled Markov Chain Monte Carlo with a 3 across-tree compositional vectors NDCH model plus 4 gamma-distributed rate categories and the LG substitution matrix (*A*), and the PhyloBayes software with a mixture model (C60) and the LG substitution matrix (*B*). The scale bars indicate the number of substitutions per site, and values near each internal branch indicate the posterior probability for that branch. Alphaproteobacterial HpsN clade sequences (and one gammaproteobacterial sequence from strain HIMB30) are marked with a blue bar, with Roseobacter clade members highlighted by the gray box and the *R. pomeroyi* DSS-3 sequence marked by a star. HpsN paralogs in Roseobacter genomes for which function has not been determined are marked with a dashed bar. The remaining sequences are candidate HpsN proteins with varying levels of bioinformatic support. Trees were rooted using a variety of histidinol dehydrogenase sequences. Locus tags are given in parentheses.

Other Supporting Information

Table S1 (DOCX)
Table S2 (DOCX)
Table S3 (DOCX)
Table S4 (DOCX)